Dynamic Benchmarking of LLMs: Moving Beyond Static Evaluation

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Abstract

Large Language Models (LLMs) are used everywhere nowadays. But their evaluations still mainly rely on static benchmarks. However, as models are trained on ever-larger amounts of internet data and either intentionally or unintentionally incorporate benchmark data, test-set leakage is becoming increasingly common, making static benchmarks less effective at assessing the true capabilities of LLMs. The industry needs better, more effective methods for evaluating LLMs that can reduce or avoid test-set contamination. This study investigates how model performance and model ranking differ when evaluated on traditional static benchmarking question sets, such as MMLU, and on more dynamic benchmark sets, such as livebench or metalinguistics benchmarks. The study evaluated four different types of LLM models from Chinese and American frontier labs: GPT-5.4, Claude Sonnet 4.6, GLM-5.1, and MiniMax M2.7 on three different types of benchmarks: static knowledge benchmark (MMLU), dynamic, contamination-limited benchmark (LiveBench), as well as a domain knowledge type expert level metalinguistics dynamic benchmark (Begus et al) The results shows a significant performance drop between the static and dynamic benchmark results. On MMLU, all four models achieved extremely high scores, ranging from 85.8% to 96.0%. However, when tested on the more dynamic benchmark, LiveBench, Claude Sonnet 4.6 remained the top-performing model with a score of 84.6%, a 10% performance drop. While all three other models scored between 53.9% and 59.6%, with an average 30%

performance drop, this shows a clear difference relative to their static benchmark performance. The metalinguistics benchmark also showed differences in what kinds of knowledge and reasoning each model was stronger at. GLM-5.1 and GPT-5.4 performed best on theoretical linguistic analysis, with accuracy scores of 62.9% and 61.8%, respectively. While MiniMax M2.7 dropped to a significantly lower score of only 37.7%. Overall, these results suggest that static benchmark scores may be inflated by test-set leakage and do not accurately reflect the LLM's true capabilities. Future LLM evaluations, especially in computational linguistics, should use more dynamic, frequently updated, and expert-level benchmarks to better measure real reasoning ability and generalization.

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1. Introduction

Large Language Models (LLMs) have become the foundamental to AI technologies and everyday life. Yet reliably evaluating their true language, comprehensive, and real-world capability remains a very hard challenge (Liang et al., 2022). As stated in Liang et al., benchmarks orient AI, they encode values and priorities that specify directions for the AI community to improve upon. Aside from that, when implemented and interpreted appropriately, they also build transparency that enables the broader community to better understand AI technology and influence its developmental trajectory (Liang et al., 2022). Generally, the industry relies on two distinct methods to measure performance: static and dynamic evaluation. Static benchmarks typically present models with simple-to-evaluate tests that measure classification accuracy on multiple-choice questions (Hendrycks et al., 2021). Conversely, dynamic and task-based benchmarking frameworks include frequently updated questions drawn from recent information sources and automatically score answers against objective ground-truth values (Jimenez et al., 2024; White et al., 2025).

Despite the prevalence of static evaluation, these traditional methodologies exhibit significant flaws that misrepresent a model's true capabilities. Because static benchmarks do not change, and their answers are published online and often used to train newer models, a model's performance on those benchmarks will be artificially inflated if it has been exposed to the test questions during pretraining. This phenomenon is known as “test set contamination” (White et al., 2025). As stated in White et al., test set contamination, in which test data from a benchmark ends up in a newer model’s training set, is a well-documented obstacle to fair LLM evaluation and can quickly render benchmarks obsolete. Additionally, because most mainstream benchmarks focus heavily on broad general knowledge or basic mathematics, they consistently fail to evaluate models' deep, expert-level knowledge within specific theoretical domains, such as formal computational linguistics. As Beguš et al. (2025a) note, while most benchmarks assess a model’s ability to produce text based on behavioral language use, they do not examine a model's metalinguistic performance. Rather than assuming a language model possesses genuine, human-like "understanding," metalinguistic evaluation serves as a rigid probe of a model's ability to extract linguistic generalizations from formal representations and explicitly generate structural analyses of language data (Beguš et al., 2025a).

In order to get a deeper understanding of these evaluation gaps, this study investigates how the performance rankings and assessed capabilities of contemporary LLMs shift when moving from traditional static evaluations to dynamic and metalinguistic frameworks. My hypothesis is that while frontier models will exhibit exceptionally high performance on static tests, these artificially inflated scores will decrease significantly under dynamic and metalinguistic conditions, where models can no longer rely on memorized training data. To test this, four state-of-the-art models—GPT-5.4, Claude Sonnet 4.6, GLM-5.1, and MiniMax M2.7—were evaluated across the static MMLU baseline, the dynamic LiveBench framework, and an expert-level metalinguistic dataset (Beguš et al., 2025b). The experimental results largely confirm this hypothesis: while all models achieved between 85.8% and 96.0% on the static MMLU, the majority experienced a severe performance drop on the dynamic LiveBench. Aside from that, the metalinguistic evaluation inverted the static performance rankings, demonstrating that while frontier models excel at surface-level pattern recognition, they consistently struggle to analyze and understand deeper structural relationships in language, such as generating hierarchical syntactic trees.

2. Background

Traditionally, the evaluation of language models has evolved alongside their increasing capabilities. Early benchmarks, such as the Winograd Schema Challenge (WSC), were designed to specifically test common sense reasoning, world knowledge, and language understanding through the interpretation of pronouns (Levesque et al., 2012). For example, a classic WSC task presents a pair of sentences: "The trophy did not fit into the brown suitcase because it was too big" versus "...because it was too small," and asks the model to correctly identify what was too big or too small—the trophy or the suitcase (Levesque et al., 2012). As models began to master individual tasks, researchers introduced aggregated benchmark suites like GLUE (Wang et al., 2018) and its more difficult successor, SuperGLUE (Wang et al., 2019), which compiled multiple natural language understanding tasks to evaluate general, multi-task capabilities.

However, as frontier models rapidly saturated these linguistically focused suites, the field shifted toward massive, broad-knowledge static benchmarks. A prominent contemporary example is the Measuring Massive Multitask Language Understanding (MMLU) benchmark, which was designed to evaluate multitask accuracy across 57 academic and professional subjects, including elementary mathematics, United States history, computer science, and law (Hendrycks et al., 2021). The MMLU benchmark evaluation relies entirely on a multiple-choice format in which models select from four static options, and aggregate performance is measured simply as the total percentage of correct selections (Hendrycks et al., 2021).

Despite the prevalence of these standard static benchmarks, they face significant theoretical criticism regarding what they actually measure. As Linzen (2020) observes, because the training and testing data for language models often originate from the very same distribution, these evaluations frequently reward models for learning superficial statistical patterns rather than demonstrating genuine, human-like generalization. Because LLMs are fundamentally optimized for next-word prediction, their computational processes—and apparent reasoning skills—depend heavily on the input and output frequency patterns present in their massive training corpora (McCoy et al., 2024). This underlying architecture creates a phenomenon known as the "embers of autoregression," which dictates that models perform significantly better on tasks and sequences they have frequently encountered during pretraining, relying on probability and memorization rather than adaptable logic (McCoy et al., 2024). Strikingly, recent research confirms that even the newest generation of frontier models explicitly optimized for complex reasoning continues to exhibit this severe sensitivity to output probability and frequency patterns.

Consequently, exceptionally high scores on static benchmarks do not necessarily indicate robust, independent problem-solving abilities; rather, they often reflect a model's statistical familiarity with highly probable text. When models are evaluated on novel or dynamic tasks that shift away from these memorized frequency distributions, their perceived competence quickly degrades.

2.1 The Shift Toward Dynamic Benchmarks

To effectively circumvent test-set leakage, researchers have developed "dynamic" benchmarking frameworks that refresh their test sets frequently (White et al., 2025). To clarify, data "contamination" does not imply that the training corpus is inherently flawed or toxic; rather, it refers to the specific, unintended overlap between a model's pretraining data and the evaluation test set. Because LLMs operate statistically to predict the next word based on frequency distributions, exposing a model to specific benchmark questions and answers during pretraining artificially increases the likelihood that it will reproduce those exact collocation patterns during testing. Dynamic evaluations, such as LiveBench, combat this by continuously introducing novel questions derived from recent information sources and scoring answers automatically according to objective ground-truth values (White et al., 2025). Because the test questions are constantly changing, the probability of the exact test queries appearing in a model's pretraining corpus is virtually eliminated. This requires the model to process entirely novel data distributions, providing a much clearer assessment of its ability to generalize to new tasks rather than its ability to rely on pre-existing statistical associations (White et al., 2025).

However, maintaining dynamic evaluations presents significant logistical challenges. Creating tests that continuously update is resource-intensive, and even newer task-based benchmarks often contain underlying components that remain relatively static (Jimenez et al., 2024). Consequently, without rigorous, continuous updates, even dynamic tests can eventually suffer from test-set leakage as their static components inevitably get scraped into the training corpora of newer models. Furthermore, despite the development of these dynamic metrics, it remains surprisingly rare in the current literature to comprehensively compare new dynamic evaluations side-by-side with older static tests under identical, strictly controlled experimental conditions (White et al., 2025).

Additionally, evaluating complex, open-ended responses—such as formal metalinguistic analyses—introduces a scoring dilemma, as these tasks lack simple objective ground truths for programmatic grading. To address this, researchers are increasingly using the "LLM-as-a-judge" paradigm, in which a highly capable language model uses predefined criteria to evaluate generated outputs (Zheng et al., 2024). While frameworks like LiveBench explicitly avoid this method by relying exclusively on objective, verifiable ground-truth values (White et al., 2025), the LLM-as-a-judge approach provides a highly scalable and relatively reliable way to score open-ended benchmarks and complex linguistic tasks that would otherwise require prohibitive amounts of expert human labor (Li et al., 2024).

2.2 Evaluating Metalinguistic Abilities in LLMs

Furthermore, within the specific domain of computational linguistics, evaluating a model's true capability requires moving beyond general knowledge trivia to assess formal metalinguistic abilities. Rather than assuming that language models possess human-like cognition or "understanding," metalinguistic evaluation serves as a rigorous probe of a model's ability to extract linguistic generalizations from formal representations. While all LLM outputs—including metalinguistic analyses—are ultimately generated through statistical next-word prediction based on collocation patterns, metalinguistic tasks require the model to explicitly generate complex, structural analyses of language data rather than merely engaging in behavioral language use. (Beguš et al., 2025a) Because most standard artificial intelligence benchmarks focus on broad trivia or basic mathematical problem-solving, they fail to adequately assess a model's capacity to maintain the simultaneous, multi-step structural tracking required for expert-level linguistic abstraction.

Comprehensive formal linguistic evaluation encompasses several highly specialized domains designed to test these generalizations. Syntactic analysis tests whether a model can accurately reproduce the structural rules of language by generating rigidly nested, hierarchical syntactic trees; (Beguš et al., 2025a) semantic analysis evaluates its capacity to formally map structural ambiguity across different sentence parses rather than predicting sequential word probabilities at face value; metaphorical analysis assesses a model's capacity to process nuanced figurative language in context; and phonological analysis examines how well a text-based system can map underlying sound architectures, such as generalizing sound patterns and applying formal phonological rules to novel data. These tasks differ significantly from standard conversational text generation because they demand strict structural formalization—such as co-indexing a trace across multiple nested clauses using LaTeX code—that pushes the boundaries of what simple frequency-based autoregression can achieve.

To address these critical evaluation gaps, this study asks the following research question: Compared with traditional static evaluation benchmarks like the MMLU (Hendrycks et al., 2021), how do dynamic evaluations like LiveBench (White et al.,

2025) and formal metalinguistic evaluations (Beguš et al., 2025a) alter the performance rankings and assessed generalization capabilities of contemporary LLMs? I hypothesize that while frontier models will exhibit high performance on static tests, the large performance gaps and inflated scores observed on static leaderboards will narrow under dynamic and metalinguistic conditions. In these uncontaminated environments, models can no longer rely heavily on the statistical familiarity of memorized training data and must instead demonstrate robust generalization to novel problems and formal structures (Beguš et al., 2025a; White et al., 2025).

3. Methods

In this study, I evaluate four state-of-the-art frontier language models to capture the most current landscape of artificial intelligence capabilities. To ensure a highly controlled and rigorous comparative analysis, these models were tested across specific subsets of static, dynamic, and metalinguistic benchmarks, with strict output parsing and automated scoring mechanisms applied uniformly across all tests.

3.1 Selected Models and Setup

The selected models provide a balanced comparative analysis across the current frontier of artificial intelligence capabilities. OpenAI's GPT-5.4 and Anthropic's Claude Sonnet 4.6 are evaluated as state-of-the-art proprietary models. GPT-5.4 is deployed as a flagship, general-purpose reasoning baseline that brings together recent advances in reasoning, coding, and agentic computer-use capabilities—meaning the model can autonomously operate computer software and execute complex workflows across different applications (OpenAI, 2026). Claude Sonnet 4.6 is a highly capable proprietary model demonstrating strong capabilities in software engineering, long-context reasoning, and autonomous, agentic knowledge work, such as independently executing multi-step research tasks (Anthropic, 2026). To provide a rigorous comparison against these proprietary systems, GLM-5.1 and MiniMax M2.7 are included as state-of-the-art open-weight Chinese models, allowing the study to directly compare closed ecosystems against open-access architectures. Built upon the GLM-5 architecture, GLM-5.1 unites agentic, reasoning, and coding capabilities to handle complex, long-horizon tasks by autonomously running experiments, reading results, and revising its own strategies over thousands of tool calls (Zeng et al., 2026; Z.ai, 2026). Finally, MiniMax M2.7 serves as an extremely efficient frontier alternative, with its documentation emphasizing a deep understanding of complex engineering systems and a capacity for recursive self-evolution—meaning the model can autonomously collect feedback and iteratively optimize its own learning processes and computational harnesses without human intervention (MiniMax, 2026).

To ensure a fair comparison, all models were tested under uniform conditions. Following recent methodological precedents in formal linguistic evaluation, the models' temperature parameters were maintained at their provider defaults, as the literature indicates that within standard ranges, temperature variation does not significantly alter the models' core problem-solving capabilities on these specific tasks (Beguš et al., 2025a). Furthermore, for models that generate explicit reasoning traces, all reasoning chain output blocks were preserved in the raw output logs for qualitative review but were entirely stripped prior to automated grading. Any test items that failed to process due to API timeouts or token exhaustion were excluded from the final tallies, and accuracy was computed strictly over the available, successfully generated responses.

3.2 Evaluation Benchmarks

To comprehensively assess these models without incurring unnecessary computational overhead in saturated task categories, this study uses highly targeted subsets from three distinct benchmarking frameworks. The Measuring Massive Multitask Language Understanding (MMLU) benchmark serves as the traditional, industry-recognized baseline for static factual knowledge and reasoning (Hendrycks et al., 2021) Rather than using the full dataset, we utilized a targeted subset of 480 items spanning eight specific categories: Abstract Algebra, College Computer Science, College Mathematics, Formal Logic, High School Computer Science, Logical Fallacies, Philosophy, and Professional Psychology. This benchmark tests models in a static multiple-choice format, requiring them to output a single letter corresponding to the correct answer. In terms of scoring, accuracy is determined via an automated regular expression (regex) extraction that isolates the first A, B, C, or D letter generated by the model, utilizing five distinct fallback patterns to account for case sensitivity and formatting variations. To test the models' general reasoning capabilities without the confounding variable of test-set contamination, I utilized a subset of LiveBench (White et al., 2025)

LiveBench combats data contamination by featuring questions that are frequently updated with recent information sources and objective ground-truth values (White et al., 2025). This study evaluated the models on a subset of 100 items covering only the Language and Reasoning categories. LiveBench presents open-ended problems requiring objective deduction, and the expected output format varies strictly by task. To illustrate, a Reasoning task might present a complex logic grid—such as a "Zebra puzzle"—in which the model must deduce a specific trait of a person from a list of constraints, outputting only the final word (White et al., 2025). Conversely, a Language task might involve a "Connections" word puzzle in which the model must sort a randomly selected list of 16 words into four thematic categories (White et al., 2025). Scoring relies entirely on objective, programmatic grading without an LLM judge. For reasoning tasks, accuracy is computed using exact-match extraction that pulls the model's final answer from explicitly enclosed tags, such as the <solution> tag. Conversely, for the language tasks, scoring relies on a frozenset-based word-group comparison. In this programmatic evaluation, the grading script converts the grouped words generated by the model into unordered mathematical sets (frozensets), allowing the script to accurately verify if the correct words were grouped together regardless of the sequence in which the model printed them, which also allows for partial credit based on official LiveBench methodologies (White et al., 2025).

Finally, to assess formal domain knowledge, we evaluated the models using the expert-level metalinguistic dataset introduced by Beguš et al. (2025b). It is important to emphasize that this framework is fundamentally different from other benchmarks: it not only tests behavioral task performance but also requires explicit theoretical analysis grounded in formal representations. It is specifically designed to test the linguistic generalizations derived from formal representations that models might have extracted during pretraining (Beguš et al., 2025a). This dataset consists of 120 items strictly divided across four categories: Ambiguity, Movement, Phonology, and Recursion. Within the Ambiguity category, models must identify whether a given sentence is syntactically ambiguous and generate hierarchical X-bar syntactic trees in LaTeX for each possible parse. For example, a model is given the sentence "The wet rodent lab has had a leak" and must formally map the structural difference between a lab for wet rodents and a wet lab for rodents. For the Movement tasks, models are required to generate an X-bar syntactic tree in LaTeX that correctly represents all instances of syntactic movement—such as mapping wh-movement and subject-auxiliary inversion in the sentence, "Who did they invite here?"—using properly co-indexed traces. In the Phonology category, models are presented with 40 surface forms from an unfamiliar language and must identify the active phonological process, explicitly describing the underlying phonemes, surface realizations, and conditioning environments. Finally, the Recursion category requires models to determine whether a sentence contains a recursive structure, identify the specific type of recursion (e.g., prepositional phrase recursion or clausal embedding), and generate the corresponding X-bar syntactic tree in LaTeX.

Overall, this benchmark tests whether models can explicitly formalize the structural rules of language, producing highly complex and open-ended expected outputs. Because these tasks lack simple objective ground truths for regex extraction, we utilized the "LLM-as-a-judge" paradigm. GPT-5.4 acted as the evaluator, using a highly specific, pre-defined rubric to grade the open-ended responses. All 120 items were fully re-judged in this pass, with scores normalized to a 0.0–1.0 scale per item to calculate the final accuracy.

4. Results

The primary objective of this study was to determine how dynamic evaluations (White et al., 2025) and formal metalinguistic assessments (Beguš et al., 2025a) affect the performance rankings and assessed reasoning capabilities of contemporary large language models compared with traditional static benchmarks such as the MMLU (Hendrycks et al., 2021). The evaluation yielded a comprehensive dataset across these three distinct testing frameworks, revealing significant performance shifts when models were deprived of contaminated training data.

4.1 Baseline Performance vs. Overall Generalization

Under traditional static testing conditions, all four frontier models exhibited exceptionally high accuracy, consistent with performance metrics frequently touted on industry leaderboards. As shown in Figure 1, on the Measuring Massive Multitask Language Understanding (MMLU) benchmark, GLM-5.1 by Z.ai established a dominant baseline with an accuracy of 96.03% (Zeng et al., 2026; Z.ai, 2026). Anthropic's Claude Sonnet 4.6 followed closely with 95.42% (Anthropic, 2026), MiniMax M2.7 achieved a highly competitive 90.62%, and OpenAI's GPT-5.4 recorded the lowest baseline accuracy of the cohort at 85.83% (OpenAI, 2026).

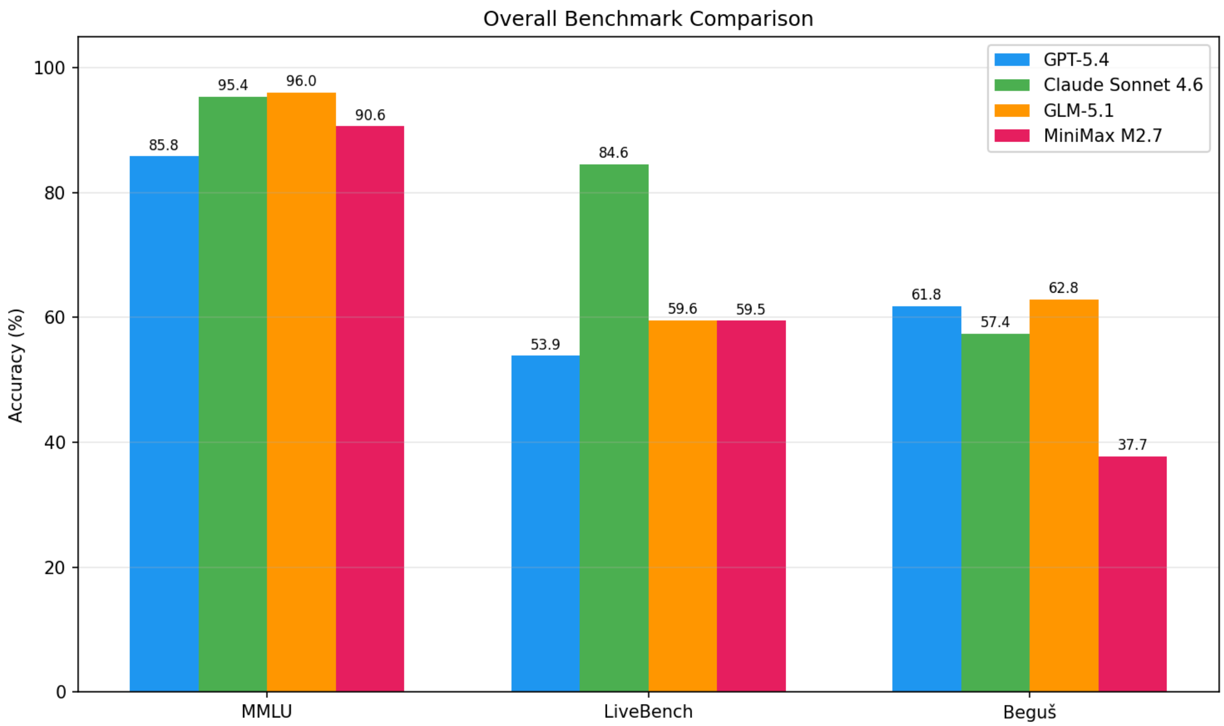

Figure 1: LLM Performance Across Evaluation Frameworks

While these high static scores might initially appear to demonstrate robust generalization across all four systems, they fail to distinguish genuine, adaptable problem-solving from the mere statistical regurgitation of a massive pretraining dataset. Indeed, calculating the models' overall average accuracy across the entire evaluation suite—which introduces the dynamic LiveBench and the open-ended formal metalinguistic tasks alongside the MMLU—strips away this statistical familiarity and reveals a much lower true capability baseline.

When all three benchmarks are averaged equally, as shown in Figure 2, Claude Sonnet 4.6 takes the overall lead at 79.14%, followed by GLM-5.1 at 72.83%. GPT-5.4’s average sits at 67.17%, while MiniMax M2.7 drops to the bottom of the cohort with a unified average of only 62.60%.

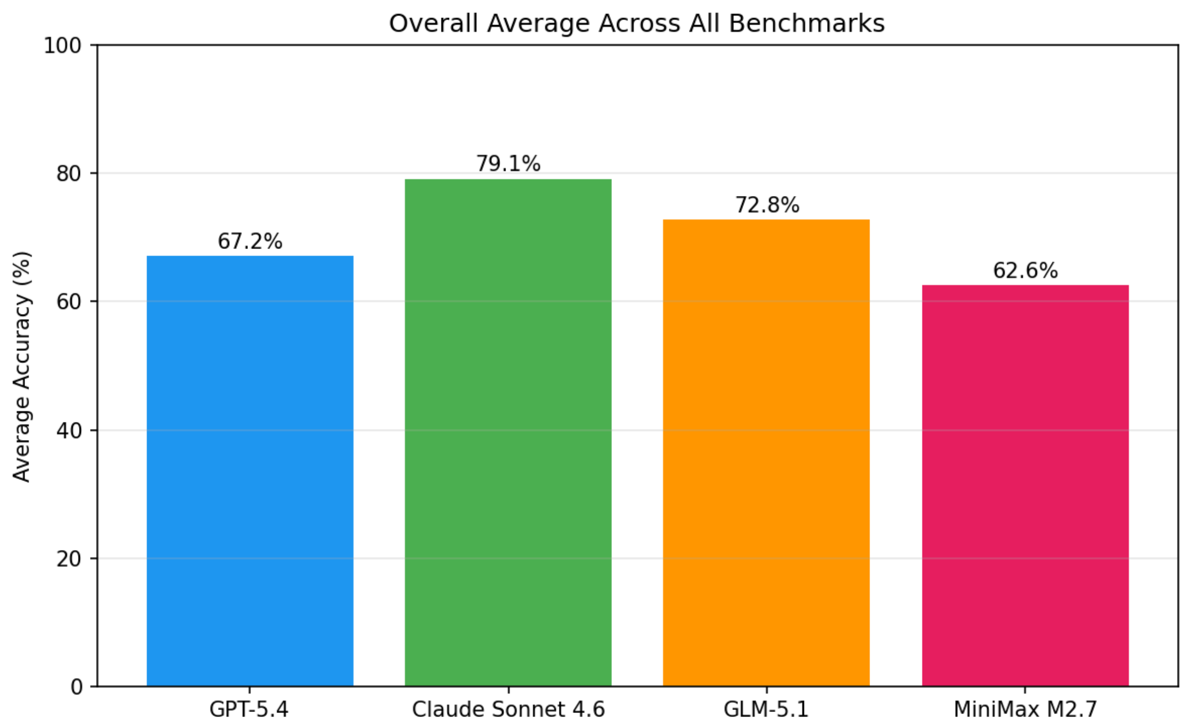

Figure 2: Overall Average Across All Benchmarks

Looking at the macro-level statistics across the individual benchmarks reveals that the high scores seen on the MMLU mask wild variations in cross-domain performance and artificial score inflation. On the continuously updated, contamination-limited LiveBench framework, the performance spread was massive. Claude Sonnet 4.6 maintained a dominant outlier accuracy of 84.58%, while the remaining three models experienced severe degradation, collapsing to 59.60% (GLM-5.1), 59.47% (MiniMax M2.7), and 53.87% (GPT-5.4) (White et al., 2025).

Furthermore, the expert-level metalinguistic dataset proved to be the most challenging environment overall. While a model's capacity to generate formal linguistic outputs is inherently tied to the volume of academic linguistics literature—such as journals or textbooks—present in its training corpus, the design of this evaluation specifically isolates a model's ability to extract structural generalizations from pure memorization. Because the test items rely on entirely novel sentences and unfamiliar or invented phonological surface forms, a model cannot simply regurgitate a memorized analysis; it must actively apply acquired theoretical formalisms to uncontaminated data. On these formal linguistic tasks, no model exceeded 63% accuracy. GLM-5.1 and GPT-5.4 inverted the static rankings, taking the lead with 62.85% and 61.81%, respectively, while Claude Sonnet 4.6 fell to 57.43%, and MiniMax M2.7 dropped to 37.71%. Ultimately, this macro-level data indicates that a high score on a traditional static benchmark does not reliably translate to a robust ability to generalize structural rules or perform expert-level theoretical analysis.

4.2 Examining the MMLU Benchmark Performance

An analysis of the isolated MMLU benchmark results initially projects an illusion of robust capability across all four systems. However, examining the granular distribution of accuracy across all benchmarks reveals that this static baseline artificially inflates the models' assessed abilities. To understand this discrepancy, we can examine the cross-benchmark heatmap (Figure 3). A heatmap is a data visualization tool that uses a color gradient to represent numerical values, allowing readers to quickly identify performance patterns across a large dataset. In this specific figure, the vertical axis lists the four evaluated models, while the horizontal axis displays the individual sub-categories for all three testing frameworks. The color scale maps directly to the models' accuracy: dark green indicates high accuracy (approaching 100%), yellow indicates moderate performance (around 40% to 60%), and red indicates severe failure (approaching 0%).

By reading the heatmap step by step from left to right, the capability drop becomes clear. First, the left-hand columns representing the static MMLU categories are universally dominated by dark green, reflecting the artificially high scores on familiar factual knowledge. However, as the chart moves to the right into the dynamic LiveBench and open-ended metalinguistic categories, the colors abruptly transition into yellows and deep reds—most notably in the Beguš Movement category, where accuracy drops to as low as 7%. This stark visual discrepancy provides undeniable evidence that static MMLU performance is not a reliable predictor of a model's ability to solve novel, real-world problems.

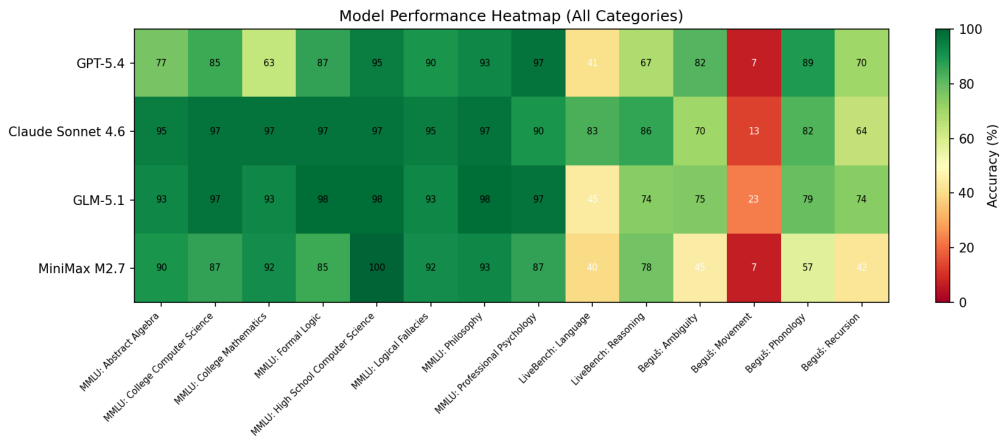

Figure 3: Accuracy Heatmap: Model x benchmark

A deeper category-level analysis of the MMLU data (Figure 4) reveals precisely which types of questions drive this score inflation. All four models performed exceptionally well on subjects that rely heavily on declarative knowledge, factual recall, and text-based memorization. For example, in the High School Computer Science category, performance ranged from 95.0% for GPT-5.4 up to a perfect 100.0% for the highly efficient MiniMax M2.7, despite it being the smallest model in the cohort (MiniMax, 2026). Similarly, in the Philosophy category, scores spanned from 93.3% for both GPT-5.4 and MiniMax M2.7 to an impressive 98.3% for GLM-5.1 (Z.ai, 2026). Professional Psychology followed this exact trend, with the cohort achieving between 86.7% and 96.7% accuracy. The near-ceiling performance in these specific areas suggests that contemporary models have successfully ingested and memorized vast quantities of standard academic curricula during their pretraining phases, enabling them to readily recognize and regurgitate answers to standard multiple-choice trivia (Hendrycks et al., 2021).

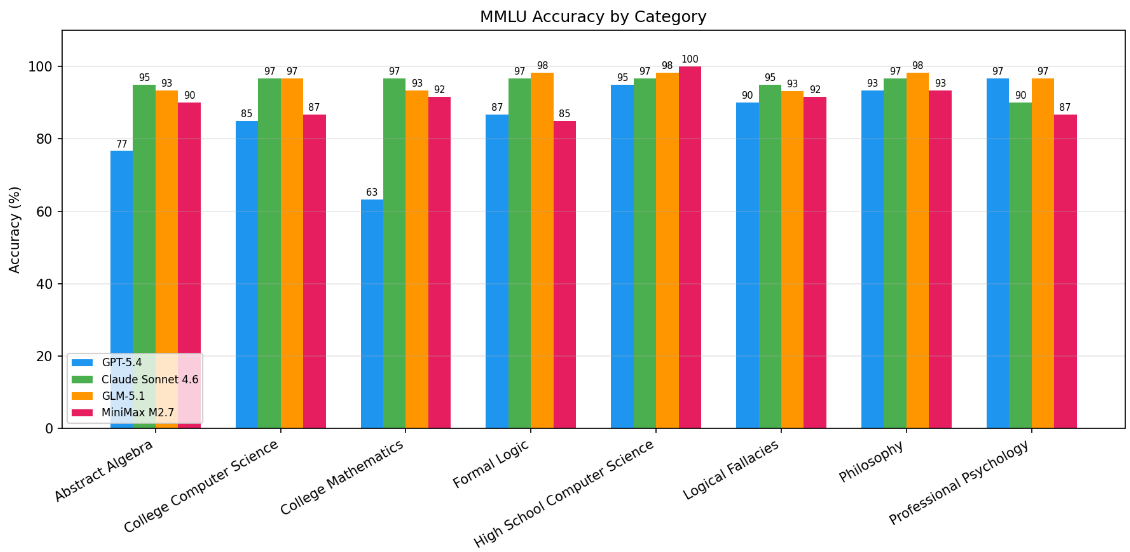

Figure 4: MMLU Accuracy by Category

Conversely, the models exhibited relative weaknesses in calculation-heavy and procedural subjects, even within this highly familiar static environment. This limitation is expected given their underlying architecture: because large language models are optimized for next-word prediction, their ability to perform mathematical computations is an emergent property rather than a systematic, rule-based process. As McCoy et al. (2024) demonstrate, model computations depend heavily on the input and output frequency in the training dataset. For example, if the standard Celsius to Fahrenheit conversion formula of f(x)=(9/5)x+32 is slightly altered to a rare variant like

$f(x)=(7/5)x+31$, model accuracy collapses. This proves that models often rely on statistical familiarity with common equations rather than applying adaptable logic.

Interestingly, OpenAI's GPT-5.4 recorded the lowest overall MMLU score at 85.8%, scoring only 63.3% in College Mathematics and 76.7% in Abstract Algebra. Meanwhile, Anthropic's Claude Sonnet 4.6 maintained highly stable accuracy (95.0% and 96.7%) across those same mathematical categories. This divergence does not necessarily indicate a fundamental difference in structural capability; rather, it highlights the impact of utilizing different pretraining corpora. Because Claude Sonnet 4.6 and GPT-5.4 are trained on distinct, proprietary datasets, Claude may have ingested a larger volume of college-level mathematics text, giving it a statistical advantage in predicting the correct token sequence for those specific problems.

Furthermore, GPT-5.4's category breakdown suggests a counterintuitive architectural phenomenon: models heavily optimized for complex, long-horizon chain-of-thought reasoning may actually underperform on straightforward, static multiple-choice questions. Because models like GPT-5.4 are structurally optimized to probabilistically generate extensive intermediate reasoning tokens before arriving at an answer, forcing them to output a single, direct multiple-choice letter disrupts their standard generation patterns, potentially introducing noise and reducing their likelihood of selecting the correct target token.

Ultimately, this granular breakdown exposes the mechanics of result inflation on traditional AI leaderboards. When a model like MiniMax M2.7 can score 100.0% on a static computer science test but immediately drops to 40.2% on LiveBench's uncontaminated language tasks, it becomes evident that the static score is heavily compromised by test-set leakage. Exceptionally high scores on the MMLU do not demonstrate an ability to generalize to novel tasks; instead, they reflect a model's statistical familiarity with a fixed dataset that was scraped into its massive internet training corpus. Therefore, relying exclusively on these inflated static benchmarks completely obscures the models' true limitations when required to process entirely novel, uncontaminated data distributions.

4.3 The Data Contamination Effect on Dynamic Benchmarks

When transitioning from the static MMLU to the continuously updated, dynamic LiveBench framework, three of the four evaluated models experienced severe, immediate performance degradation. As shown in Figure 5, GLM-5.1's accuracy dropped significantly from 96.0% on the static baseline to 59.6% on LiveBench (a 36.4% drop). Similarly, MiniMax M2.7 fell from 90.6% to 59.5% (a 31.2% drop), and GPT-5.4 dropped from 85.8% to 53.9% (a 31.9% drop). While this massive decline is highly consistent with the data-contamination hypothesis—as LiveBench continuously introduces novel questions from recent sources, preventing models from relying on test-set leakage—it is important to recognize that contamination is not the sole variable at play. The performance drop could also be attributed to differences in task format, difficulty, and scoring mechanisms between the two benchmarks. Furthermore, as McCoy et al. (2024) observe, large language models are fundamentally optimized for next-word prediction and are therefore highly sensitive to the frequency patterns present in their training data. Because dynamic evaluations like LiveBench introduce entirely novel problem distributions, they shift models away from the high-frequency collocation patterns they rely on during static testing, which can severely degrade a model's predictive accuracy even in uncontaminated settings.

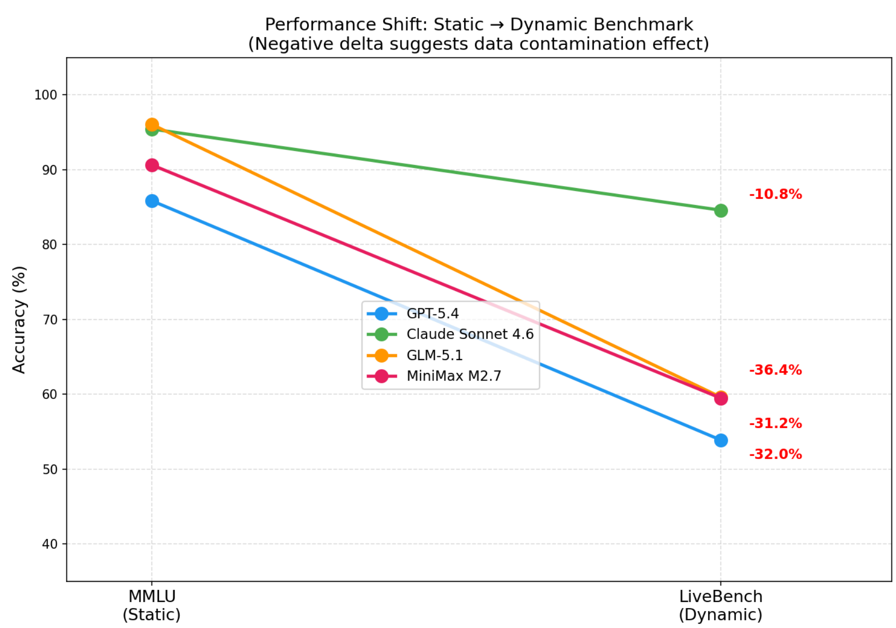

Figure 5: Static to Dynamic Benchmark Performance Shift

Conversely, Anthropic's Claude Sonnet 4.6 emerged as a significant outlier, dropping only slightly from 95.4% to a dominant 84.6% on the dynamic tasks. Although this still represents a 10.8% decrease, maintaining an accuracy above 84% on novel problems demonstrates a much higher degree of genuine, uncontaminated generalization.

A deeper categorical analysis of the LiveBench results reveals distinct capability footprints among the models. As shown in Figure 6, while Claude Sonnet 4.6 maintained high accuracy across both the Reasoning (86.0%) and Language (83.2%) subcategories, the remaining three models exhibited a stark divide. GPT-5.4, GLM-5.1, and MiniMax M2.7 remained relatively competitive on pure objective reasoning tasks, scoring between 67.3% and 77.6%. However, their performance collapsed on the Language tasks—which include lateral-association puzzles like word grouping—with all three models scoring between 40.2% and 44.9%. To illustrate, a LiveBench Language task might present a "Connections" puzzle in which the model must sort a random list of 16 words into four thematic categories, such as grouping "ant, drill, island, and opal" because they can all follow the word "fire".

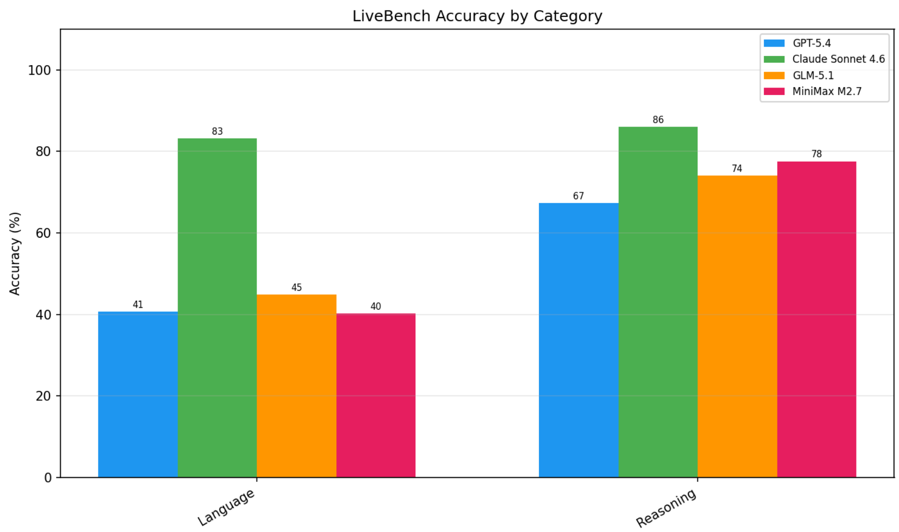

Figure 6: LiveBench Accuracy by Category

This granular breakdown suggests that models optimized to forcibly generate extensive, step-by-step chain-of-thought tokens may excel in sequential logic but struggle significantly with lateral association and open-ended linguistic deduction. Because language understanding tasks frequently rely on immediate, holistic pattern recognition rather than step-by-step mathematical derivation, forcing a model to generate long chains of intermediate tokens can actually disrupt its processing. As Beguš et al. (2025a) demonstrate, applying explicit, step-by-step reasoning requirements to certain foundational linguistic tasks can actively decrease model performance. Generating excessive reasoning tokens for a task that requires direct pattern matching introduces statistical noise into the context window, ultimately reducing the probability that the model will output the correct target sequence. Ultimately, the resulting negative performance delta confirms that traditional static leaderboards heavily overstate a model's ability to generalize, masking the extent to which systems rely on statistical familiarity with memorized text patterns rather than adaptable problem-solving.

Ultimately, the resulting negative performance delta confirms that traditional static leaderboards heavily overstate independent reasoning skills, masking the extent to which models rely on statistical familiarity with memorized test content.

4.4 Metalinguistic Reasoning and Shifts in Model Rankings

The evaluation of formal metalinguistic abilities further disrupted the initial performance rankings established by the static MMLU. In this uncontaminated, expert-level domain introduced by Beguš et al. (2025b), the model rankings are inverted. As shown in Figure 7, GLM-5.1 and GPT-5.4 demonstrated superior capability in generating formal linguistic abstractions, achieving the highest overall accuracies at 62.8% and 61.8%, respectively. This indicates that generating explicit chain-of-thought tokens actively assists models in processing deep theoretical formalizations. Unlike lateral association tasks, where generating unnecessary intermediate tokens introduces statistical noise, formal metalinguistic tasks—such as tracking co-indexed traces across a nested LaTeX syntax tree—require rigorous, multi-step structural tracking. In these strictly sequential scenarios, chain-of-thought generation actively assists the model through a process of "self-conditioning". By breaking down a complex formalism and explicitly producing intermediate tokens, the model creates an expanded, highly relevant context window to condition its subsequent predictions. This sequential generation significantly increases the probability of the model maintaining structural integrity throughout the final output. Claude Sonnet 4.6, despite its dominance on LiveBench, fell to third place with 57.4%, while MiniMax M2.7 struggled significantly with expert-level abstractions, scoring only 37.7%.

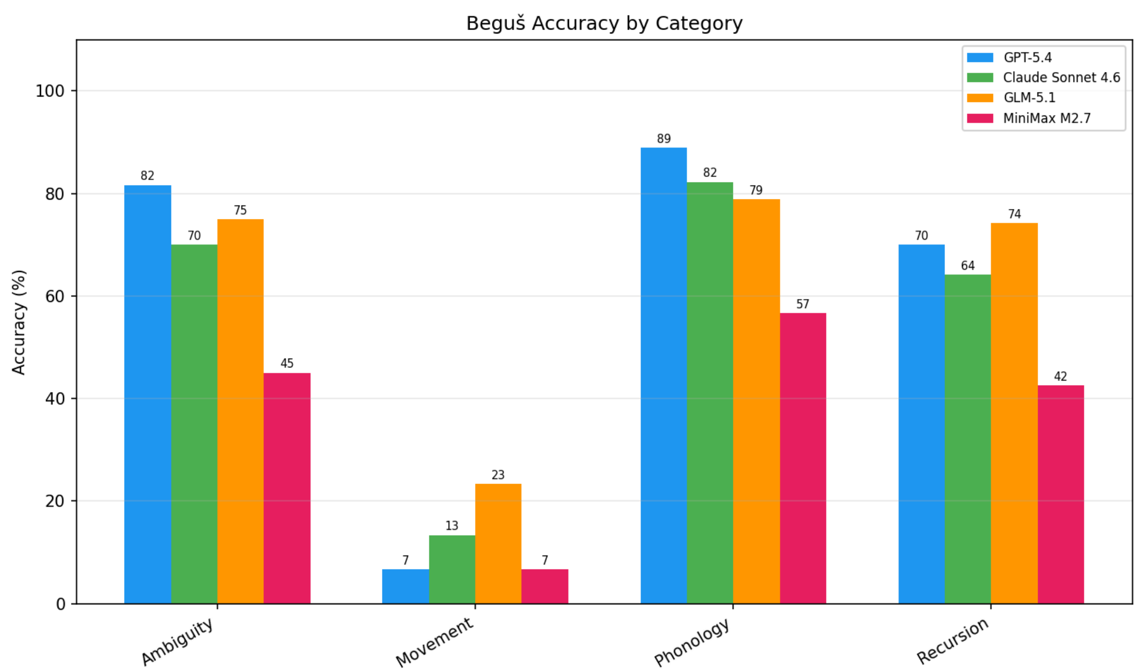

Figure 7: Beguš Accuracy by Category

A granular breakdown of the metalinguistic categories exposes the specific boundaries of current AI architectures. The models generally succeeded on the

Phonology tasks (which test the ability to identify phonological processes and describe underlying phonemes and environments), with GPT-5.4 peaking at 88.9%, and on the Ambiguity and Recursion tasks, where top models reliably scored between 64.2% and 81.7%. However, the Movement category proved to be a near-impossible structural hurdle. This category tests the ability to formally map complex syntactic movement operations, such as wh-movement or topicalization, and draw syntactic trees. On these tasks, accuracy collapsed universally: GLM-5.1 scored the highest at only 23.3%, while both GPT-5.4 and MiniMax M2.7 scored a mere 6.7%. To determine whether this collapse is simply due to a lack of linguistic research in the training corpora, it is necessary to compare the task constraints. Because models succeeded at formalizing abstract phonological rules using specific feature matrices, it is evident that advanced computational linguistics data is present in their pretraining. The discrepancy instead arises from the mechanics of autoregression: generating LaTeX trees with properly co-indexed traces across nested clauses requires simultaneous, multi-step structural tracking that standard next-word prediction cannot easily maintain. Phonological analysis, conversely, relies on more localized, sequential pattern matching.

Ultimately, these empirical results are highly consistent with the study's central hypothesis: the massive performance gaps and inflated scores seen on static leaderboards shrink dramatically under dynamic and metalinguistic conditions. However, this performance degradation cannot be exclusively attributed to the removal of test-set contamination. As McCoy et al. (2024) observe, because LLMs operate via statistical next-word prediction, their accuracy is fundamentally tied to the input and output frequency patterns of their training data. Exposing these models to entirely novel environments not only eliminates their ability to rely on memorized test sets, but it also forces them to generate low-probability, out-of-distribution sequences. Consequently, a model's proficiency in recalling high-frequency static trivia does not reliably translate into an ability to generalize to novel tasks, nor does it guarantee the capacity required to generate complex theoretical linguistic formalizations.

5. Discussion

The empirical findings of this study provide compelling evidence that traditional, static evaluation metrics fundamentally misrepresent the ability of contemporary large language models to generalize to novel tasks. By directly comparing model performance across static baselines, continuously updated dynamic frameworks, and expert-level metalinguistic tasks, this research contextualizes the limitations of standard benchmarks within the theoretical framework proposed by Linzen (2020). Because the training and testing data for static tests like the MMLU frequently originate from the very same distribution, these benchmarks artificially inflate scores by rewarding models for learning superficial statistical patterns rather than demonstrating robust, human-like generalization (Linzen, 2020). The severe performance degradation observed on both

LiveBench and metalinguistic tasks demonstrates what happens when models are forced outside this familiar distribution. In these dynamic, uncontaminated environments, the effects of test-set leakage (White et al., 2025) and high-frequency collocation patterns are stripped away, exposing a model's true capability limitations. Ultimately, these results underscore the critical need to transition to continuous, out-of-distribution, and domain-specific evaluations to accurately measure true progress in artificial intelligence (Beguš et al., 2025a; White et al., 2025).

5.1 The Illusion of Static Competence

The most striking observation from the dataset is the precipitous performance degradation that most models experienced when transitioning from the static MMLU benchmark (Hendrycks et al., 2021) to the dynamic LiveBench framework (White et al., 2025). While GLM-5.1 achieved a near-perfect 96.0% on the MMLU, its accuracy fell drastically to 59.6% on LiveBench. Similar severe performance drops were observed for MiniMax M2.7 (which fell from 90.6% to 59.5%) and GPT-5.4 (which fell from 85.8% to 53.9%). Notably, Anthropic's Claude Sonnet 4.6 emerged as a significant outlier, maintaining a robust 84.6% on the dynamic LiveBench tasks after scoring 95.4% on the static MMLU baseline. For models that experienced this massive negative delta, the precipitous decline is highly consistent with the hypothesis that exceptionally high scores on established static benchmarks are largely an artifact of test-set leakage

(White et al., 2025). However, this drop cannot be attributed exclusively to contamination; shifts in task format, evaluation difficulty, and objective scoring mechanisms between the two benchmarks also contribute to this performance difference.

Because frontier models ingest vast portions of the internet during pretraining, static multiple-choice questions primarily measure statistical familiarity and memorization rather than robust generalization (Hendrycks et al., 2021). Dynamic evaluations like LiveBench successfully mitigate this illusion of competence by continuously refreshing their prompts, preventing models from relying on high-frequency pretraining data. This provides a far more realistic assessment of a model's ability to generalize to uncontaminated, novel distributions under real-world conditions (White et al., 2025).

The stark contrast between Claude Sonnet 4.6 and the rest of the cohort further underscores that while statistical familiarity can artificially inflate static leaderboards, genuine and adaptable generalization to novel tasks remains a distinct, harder-to-achieve capability.

5.2 Expert-Level Metalinguistic Capabilities

Beyond the static-dynamic divide, the evaluation of formal metalinguistic abilities revealed significant disparities in how models generate deep linguistic abstractions.

Standard benchmarks rarely test expert-level tasks like building hierarchical syntactic trees, formalizing semantic ambiguity, or applying generalized phonological rules to novel words (Beguš et al., 2025a). These metalinguistic tasks are highly complex because they require a model to explicitly formalize the structural dependencies of language itself, rather than merely generating highly probable conversational tokens based on behavioral language use (Beguš et al., 2025a). As established, language models do not possess human-like cognition or genuinely "reason" about syntax; rather, their outputs are driven entirely by statistical next-word prediction (McCoy et al., 2024). Given that frontier models ingest vast quantities of academic literature and textbooks during pretraining, one might initially expect them to readily output standard linguistic analyses by relying on high-frequency patterns. However, successfully generating a novel hierarchical tree in LaTeX requires maintaining simultaneous, multi-step structural tracking, pushing basic autoregression to its limits.

In this uncontaminated, expert-level domain, the performance rankings inverted from the static baseline. GLM-5.1 demonstrated the strongest ability to extract and apply these formalisms, achieving the highest overall accuracy at 62.85% (Z.ai, 2026). It was closely followed by OpenAI's GPT-5.4, which achieved a score of 61.81% (OpenAI, 2026). In contrast, Anthropic's Claude Sonnet 4.6 fell to third place at 57.43% (Anthropic, 2026), and the highly efficient MiniMax M2.7 struggled significantly with the theoretical abstractions, scoring only 37.71% (MiniMax, 2026).

This shift in rankings indicates that a model's capacity to recall general-knowledge trivia or execute standard programming tasks does not inherently translate into a capacity to successfully generate complex theoretical linguistics (Beguš et al., 2025a). The ability to systematically apply formal linguistic frameworks to novel, unobserved language data remains a distinct and highly challenging computational task for artificial neural networks (Beguš et al., 2025a). While frontier models like GLM-5.1 and GPT-5.4 show promising early signs of generating these abstractions, the overall scores—all remaining well below 70%—suggest that computational linguistics remains a formidable frontier for AI development.

Ultimately, to accurately gauge the progression of artificial intelligence, researchers must continue to pivot away from static trivia and toward dynamic, expert-calibrated evaluations that definitively separate statistical familiarity with memorized text from robust generalization to novel, complex structures.

5.3 Granular Analysis of Metalinguistic Performance

A closer examination of the models' performance across the specific subtasks within the metalinguistic dataset reveals a stark contrast between surface-level categorization and deep structural formalization. The granular data demonstrates that contemporary LLMs are highly proficient at identifying linguistic phenomena when asked direct, descriptive questions, but they consistently fail when required to generate the underlying theoretical architectures of those phenomena (Beguš et al., 2025a). Rather than providing evidence of genuine "reasoning" or "thought," these results suggest that models are simply reflecting back the text-based descriptive analyses that are highly prevalent in their published pretraining corpora. Breaking these results down by question type exposes the specific statistical and architectural boundaries of current frontier models.

The models were most likely to succeed on subtasks requiring basic identification or feature extraction. Across the structural ambiguity and recursion prompts, all four models frequently output the correct categorization. For example, when presented with the sentence, "The wet rodent lab has had a leak," the models correctly articulated the two surface-level meanings: a lab for rodents that is currently wet, versus a lab specifically for wet rodents. Similarly, they reliably categorized NP-internal modifier attachment ambiguity in sentences like "The new energy theorem confuses me," correctly identifying whether "new" modified "energy theorem" or "new energy" modified "theorem". This proficiency extended to the phonology domain. When given 40 surface forms from an unknown language, models reliably identified the input and output phonemes, such as correctly noticing that an underlying voiceless stop becomes pre-glottalized, or that an oral vowel surfaces as a nasal vowel. These successes confirm that LLMs are highly capable of basic pattern recognition, precisely because similar descriptive, text-based explanations of language are abundant in their pretraining data (Beguš et al., 2025a).

Conversely, the models were least likely to succeed when tasked with translating these surface-level observations into formal theoretical frameworks. The lowest performance across all models was observed in the syntax tree generation and X-bar movement tasks. In this domain, the "Movement" category proved nearly impossible; GLM-5.1 scored the highest at only 23.3%, while GPT-5.4 and MiniMax M2.7 collapsed to a mere 6.7%. While one might assume that this failure occurs because syntactic trees rely on symbolic visual structures, such as drawn lines, the experimental design controlled for this variable. The models were specifically evaluated on their ability to generate text-based LaTeX forest code and were explicitly instructed not to write code that draws syntactic arrows. Instead, the task required models to correctly place text-based traces and mathematically co-index them with their antecedents across a nested hierarchy. For instance, when tasked with mapping the wh-movement and subject-auxiliary inversion in the sentence, "Who did they invite here?", models like MiniMax M2.7 and Claude Sonnet 4.6 frequently failed to place the origin trace in the exact correct complement or specifier node within the rigidly nested LaTeX bracket structure. Furthermore, while differences in training data volume—such as one model ingesting more LaTeX linguistics papers than another—might explain minor score variations, the universal performance collapse across all four models indicates a fundamental architectural limitation. While models can verbally output movement descriptions by relying on high-frequency collocation patterns, tracking simultaneous, long-distance structural dependencies across nested brackets fundamentally disrupts their next-word prediction algorithms (McCoy et al., 2024).

5.4 Success in Deep Structural Formalization: Phonological Rules

Although the models struggled universally to generate complex hierarchical syntax trees, the top frontier systems demonstrated remarkable proficiency in phonology. GPT-5.4, Claude Sonnet 4.6, and GLM-5.1 achieved high accuracies of 88.9%, 82.2%, and 78.9%, respectively, on the phonological tasks. When required to extract input and output sounds and formalize the conditioning environment from novel surface forms, these models successfully utilized standard natural-class notation. For example, when tasked with formalizing a rule where oral vowels become nasalized adjacent to nasal consonants, models like GLM-5.1 and GPT-5.4 did not revert to plain English or simply list the affected phonemes; instead, they accurately abstracted the raw segments into formal feature matrices, generating precise rules such as [+syllabic]→[+nasal]/_[+consonantal,+nasal].

However, the ability to generate these specific formalisms is not yet universal across all architectures. The highly efficient MiniMax M2.7 model struggled significantly in this domain, scoring only 56.7% and frequently failing to formalize the conditioning environments consistently. This divergence does not indicate a difference in underlying "metacognitive" ability or reasoning; rather, it likely reflects differences in the volume and composition of the models' respective pretraining corpora. Because MiniMax M2.7 is an efficiency-focused, smaller-parameter architecture, it is highly probable that its training dataset contained a significantly smaller volume of advanced academic linguistics literature than the massive, trillion-token corpora ingested by GPT-5.4 or Claude Sonnet 4.6. Consequently, MiniMax M2.7 lacks the robust, high-frequency statistical representations of phonological rule notation required to reliably map abstract phonetic features into formal matrices.

5.5 Mechanisms of Failure and Pathways for Improvement.

This discrepancy highlights a fundamental limitation in current AI architectures: while LLMs readily output high-probability tokens to mimic the behavioral use of language and identify surface linguistic patterns, they struggle to apply rigid, abstract linguistic formalisms to novel, uncontaminated data. To clarify, this failure is not merely an artifact of evaluating syntactic trees; rather, syntactic trees serve as a rigorous diagnostic tool that exposes a broader computational vulnerability. Because models rely heavily on token-level statistics, translating non-linear, hierarchical relationships into a linear token stream—such as co-indexing a trace across multiple nested clauses in

LaTeX—requires simultaneous, multi-step structural tracking that standard autoregressive generation cannot easily maintain.

To improve performance in these expert-level domains, future model development must move beyond standard text corpora and actively incorporate rigorous, domain-specific formalisms during pretraining or supervised fine-tuning. Furthermore, improving performance on tasks requiring rigid structural formalization may require integrating neuro-symbolic approaches. In such frameworks, the language model generates an initial probabilistic sequence, but interfaces with an external symbolic parser or formal grammar checker to programmatically verify the structural integrity of its generated LaTeX code and phonological rules before outputting a final answer.

6. Conclusion

The empirical results of this study demonstrate that while contemporary large language models continually achieve exceptionally high scores on traditional leaderboards, these static metrics frequently misrepresent their ability to generalize to novel tasks. While these artificially inflated scores are highly consistent with the effects of data contamination (White et al., 2025), assuming a strictly causal relationship between test-set leakage and the performance drop observed on dynamic frameworks would overlook other critical factors. When evaluated under the continuously updated framework of LiveBench, the accuracy of most frontier models (GLM-5.1, MiniMax M2.7, and GPT-5.4) plummeted. This performance degradation can also be attributed to differences in task format, evaluation difficulty, and objective scoring mechanisms between the two benchmarks. Furthermore, as McCoy et al. (2024) observe, large language models are fundamentally sensitive to the frequency patterns present in their training data. While the robust performance of Anthropic's Claude Sonnet 4.6 on these dynamic tasks proved that uncontaminated generalization is achievable, the severe performance degradation across the rest of the cohort indicates that much of the assessed capability of modern AI systems stems from statistical familiarity with high-frequency collocation patterns rather than robust adaptability to out-of-distribution data.

Furthermore, the evaluation of expert-level domain knowledge underscores the specific architectural and statistical boundaries of current AI systems. Within the domain of computational linguistics, the data reveal a stark contrast between surface-level pattern recognition and deep hierarchical formalization (Beguš et al., 2025a). While all models could adequately categorize general linguistic phenomena and the top frontier models successfully formalized abstract phonological rules using feature matrices, they universally failed to generate complex, multi-step hierarchical syntactic trees (Beguš et al., 2025a). This indicates that while LLMs excel at generating high-probability tokens to mimic behavioral language use and process localized abstractions, they still lack the simultaneous, multi-step structural tracking mechanisms necessary to systematically generate rigidly nested, complex formal architectures.

Ultimately, these findings carry significant implications for the future of artificial intelligence development. As long as the industry relies on static tests like the MMLU, true computational progress will remain obscured by an illusion of competence. Moving forward, the field must fundamentally shift its evaluation and training paradigms. To accurately gauge robust generalization, developers must pivot toward dynamic, continuously updated evaluation frameworks that systematically minimize test-set leakage and test models on out-of-distribution distributions. Furthermore, future training methodologies must transcend broad trivia and prioritize rigorous, domain-specific formalizations, ensuring that next-generation models possess the computational architecture required to reliably extract and apply expert-level theoretical abstractions (Beguš et al., 2025a).

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